

**PDAN8411- Programming for
statistics module**

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**Report for formative task
one**

Table of Contents

1. Introduction	3
2. Data Cleaning and Preparation	3
3. Exploratory Data Analysis	3
4. Feature Selection.....	4
5. Model Training and Performance	4
6. Model Interpretation	4
7. Limitations	4
8. Final Analysis and Conclusion	4
9. Bibliography	5

1. Introduction

The aim of this analysis was to decipher whether customer traits could be utilised to forecast medical insurance prices using multiple linear regression. The dataset contained information on age, sex, BMI, number of children, smoking status, region, and medical charges.

2. Data Cleaning and Preparation

The dataset first contained 1338 observations and 7 elements. The data types were relevant for analysis, with numerical elements including age, BMI, children and charges, and categorical variables including sex, smoker and region. No missing values were found. One replicate observation was found and removed, resulting in a final dataset of 1337 observations. The categorical variables were changed into numerical draft variables to enable them to be utilised in the regression model. The target variable was charges, while the final variables were thought of as potential forecasters

3. Exploratory data analysis

The descriptive statistics demonstrated that customer age varied from 18 to 64 years of age, while the BMI varied from 15.96 to 53.13. Medical charges differed greatly, ranging from 1121.87 to 63 770.43 dollars. The mean medical charge was approximately 13 279.12 dollars. The distribution of medical charges was positively skewed, showing that most findings had lower charges while a smaller number of consumers had significantly higher charges. The analysis also showed that age, BMI, and smoking status were essential factors related to medical charges. The categorical elements illustrated that the dataset had 675 males and 662 females. There were 1063 non-smokers and 274 smokers. The four geographical areas were adequately reflected, although the Southeast region had the biggest number of observations.

4. Feature Selection

Feature selection was executed using p-values from an Ordinary Least Squares regression model, utilising a 5% significance level. Age, BMI, number of children, smoking status, Southeast region, and Southwest region were statistically substantial and were chosen for the last model. Sex and Northwest region were not statistically important and were left out. Sex had a p-value of 0.698, and the Northwest region had a p-value of 0.464. Smoking status was very significant with a coefficient of 23 850 dollars, showing a potent relationship with medical charges.

5. Model Training and Performance

The chosen variables were split into 80% training data and 20% testing data. This outcome is shown in 1069 observations and is being utilised for training and 268 for testing. A multiple linear regression model was then trained utilising the training data.

Metric	Result
R-squared	0.8069
Adjusted R-squared	0.8024
MAE	\$4175.84
RMSE	\$5957.47

R squared shows that approximately 80.69% of the differences in medical charges were explained by the chosen forecasters. The fixed R-squared of 0.8024 displays that the model continued to explain a significant part of the variation after explaining the number of predictors. The MAE of 4175.84 dollars means that the models forecasting did not meet the actual charges by approximately 4176 dollars on average. The RMSE of 5957.47 dollars was higher because RMSE provides a larger weight to bigger forecasting errors. These values were chosen because they give a total measure of explanatory power and a piece of evidence of the size of the prediction errors.

6. Limitations

The dataset has a significant number of limitations. Beginning with the fact that it is a dataset from the United States of America, while the proposed application involves a South African medical organisation. Therefore, the outcomes may not adequately identify South African customers and their traits. Secondly, the dataset comprises a limited number of predictors, which means that the other factors influencing medical costs are not captured. The distribution of charges is positively skewed, with high-cost observations leading to more prediction errors. Lastly, statistical importance does not imply causation. Therefore, the model must be seen as a predictive proof of concept instead of a complete pricing structure.

7. Final analysis and conclusion

The outcomes displayed that multiple linear regression gives a necessary method for forecasting medical insurance charges. The model achieved an R-squared of 0.8069,

demonstrating that the chosen customer traits explained a significant part of the variation changes. Smoking status was the most potent predictor in the regression analysis while age, BMI, and number of children were also statistically important. Some regional variations were significant, while sex and the Northwest region did not provide any significance as shown by the feature-selection criterion. Overall, the model gives substantial proof of concept for utilizing customer traits to approximate medical costs. However, more testing with the South African medical scheme data and further health and claims-related variables would be important before the model could be used for actual medical-cost or pricing choices.

8. Bibliography

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